



Utilization of Gojek Features to Support Digital Transformation in The Transportation Sector in The Digital Era

Group 3

Names : Arla Bukit Elbirun
 Arlie Fabian
 Farisa Aulia Sudrajat
Class : 1CC7

CEP CCIT
FAKULTAS TEKNIK UNIVERSITAS INDONESIA

2025

PROJECT ON

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Developed by

- 1. Arla Bukit Elbirun**
- 2. Arlie Fabian**
- 3. Farisa Aulia Sudrajat**

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Batch Code : 1CC7

Start Date : October 1, 2025

End Date : October 20, 2025

Name of Faculty : Mrs. Indah Ayu Yuliani, S.T., M.M & Mr. Hary Prasetyo Rosianto, S.Tr.T.

Names of Developer :

1. Arla Bukit Elbirun
2. Arlie Fabian
3. Farisa Aulia Sudrajat

Date of Submission : October 20, 2025

ACKNOWLEDGEMENT

First of all, thanks to Allah SWT because of the help of Allah, author finished writing the paper entitled “Utilization of Gojek Features to Support Digital Transformation in The Transportation Sector in The Digital Era” right in the calculated time.

The purpose in writing this paper is to fulfil the assignment that given by Mrs. Indah Ayu Yuliani, S.T., M.M and Mr. Hary Prasetyo Rosianto, S.Tr.T. as lecturer in Information Technology Major.

In arranging this paper, the writer trully get lots challenges and obstructions but with help of many indiviuals, those obstructions could passed. Author also realized there are still many mistakes in process of writing this paper.

Because of that, the author says thank you to all individuals who helps in the process of writing this paper. Hopefully Allah replies all helps and bless you all. The writer realized this paper still imperfect in arrangement and the content. Then the author hope the criticsm from the readers can help the writer in perfecting the next paper. Last but not least hopefully, this paper can helps the readers to gain more knowledge about information technology major.

BACKGROUND

The development of information and communication technology has changed various aspects of people's lives, including in the transportation and service sectors. In this digital age, innovations have emerged that utilize technology to provide solutions to problems faced by people in their daily lives. One such innovation is the emergence of technology-based applications that offer online transportation services and various other supporting services.

Indonesia, as a country with a population of more than 270 million people, faces various challenges in the transportation sector, such as traffic congestion, limited access to adequate public transportation, and difficulties for the community in accessing various daily services efficiently. These conditions have led to the emergence of digital platforms that can bridge the needs of the community with service providers.

Gojek, founded in 2010 by Nadiem Makarim, emerged as an innovative solution that combines technology with conventional transportation services. Starting as a call center-based motorcycle taxi service, Gojek transformed into a super app in 2015 that not only provides transportation services, but also offers goods delivery, food ordering, digital payments, and various other services on a single integrated platform.

Gojek's presence has had a significant impact on Indonesia's economy. The platform has created jobs for millions of driver-partners and merchants, driven the digital transformation of MSMEs, and increased financial inclusion through GoPay services. In addition, Gojek has also changed consumer behavior in accessing services and conducting transactions digitally.

However, behind its success, Gojek also faces various challenges, ranging from fierce competition with competitors, government regulations, to issues regarding the welfare of driver partners. This makes Gojek an interesting subject to study from various perspectives, including its business model, socio-economic impact, marketing strategies, and the technology it uses.

Based on this background, this paper will analyze the features of Gojek that are practically utilized by consumers.

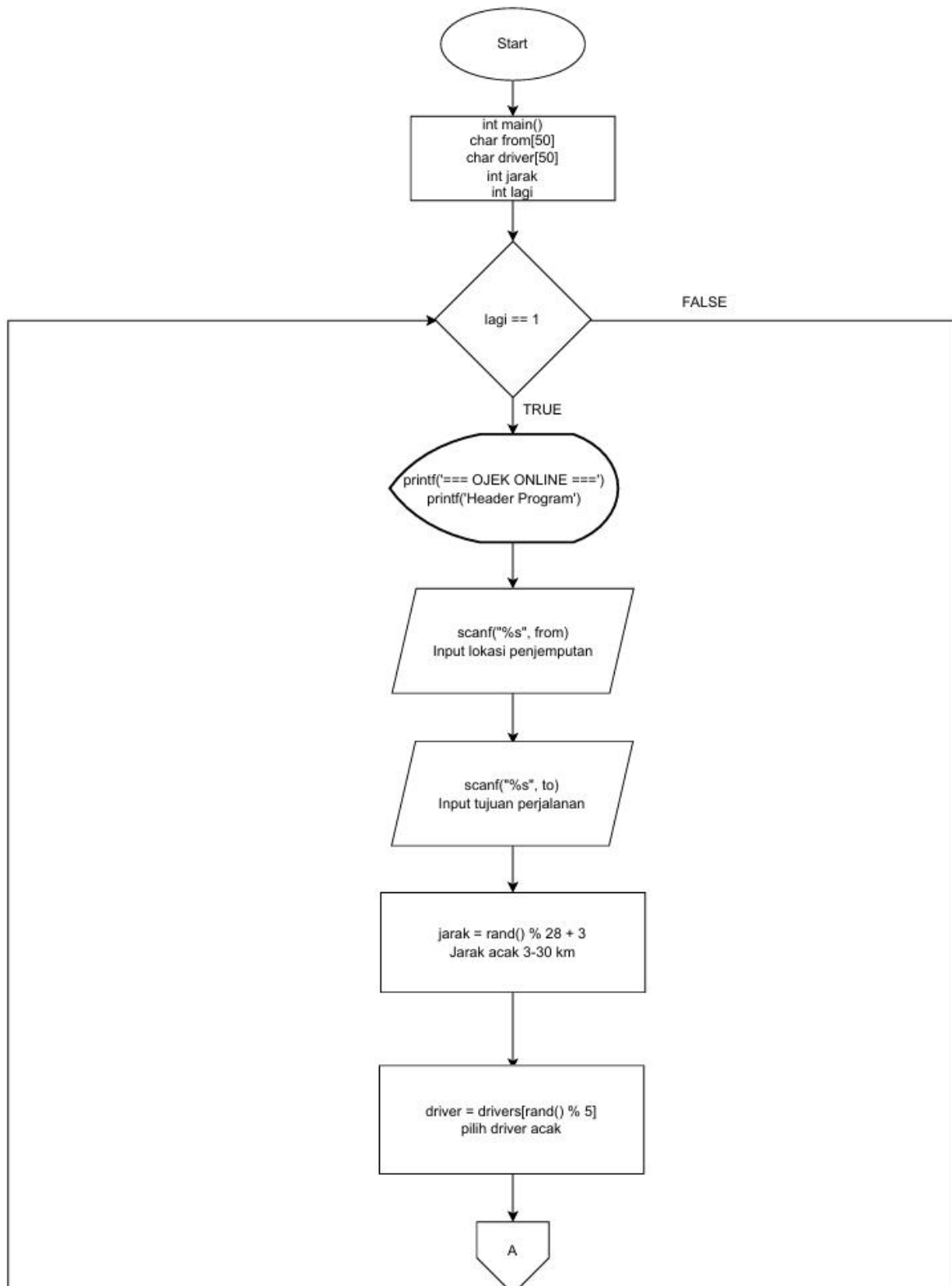
SYSTEM ANALYSIS

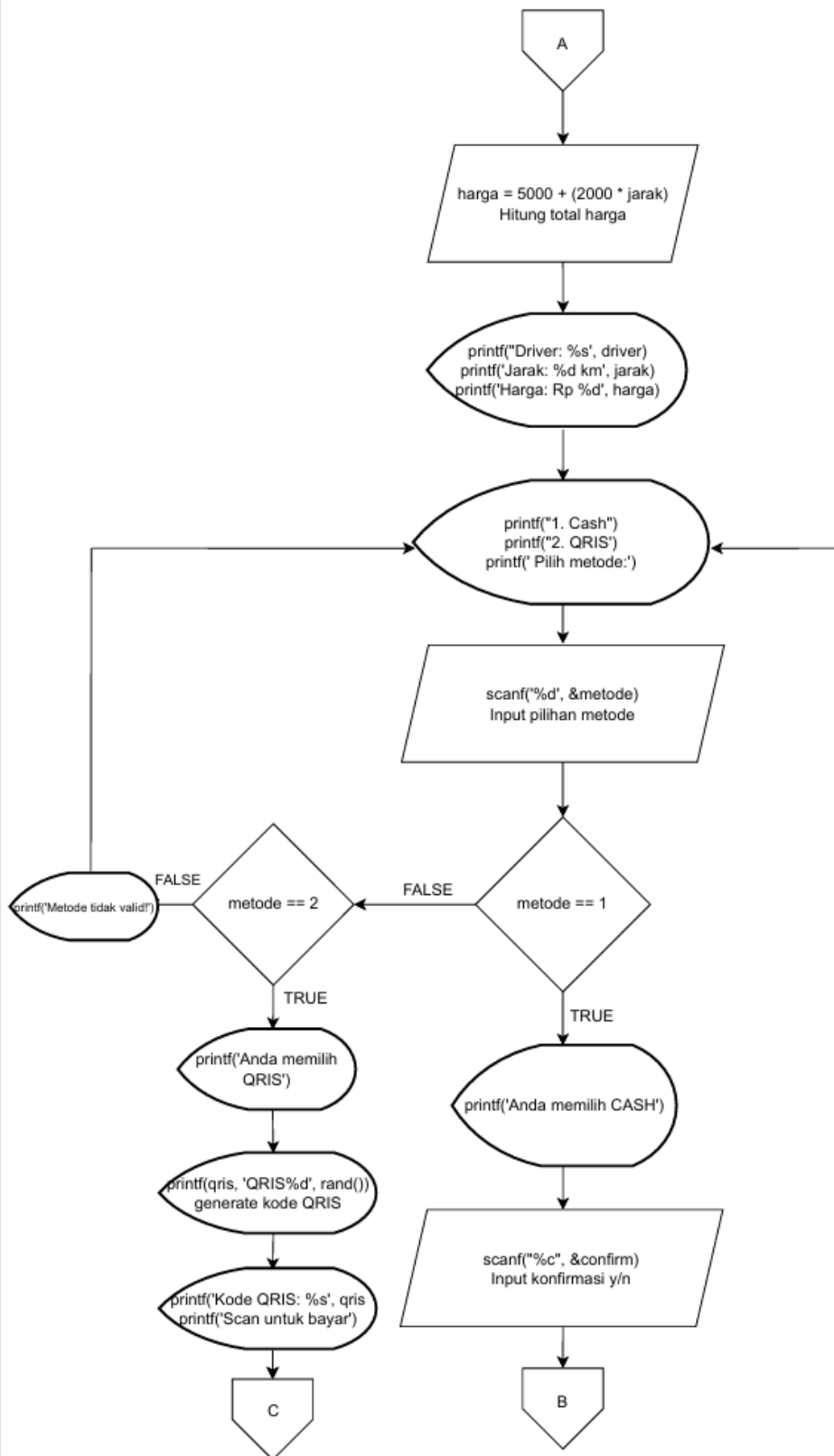
Gojek, through its GoRide service, provides an app-based online transportation booking system that connects users with motorcycle drivers. This system works in real time, utilizing GPS to determine the location of the nearest passengers and drivers, as well as algorithms to match the two efficiently. This process ensures short waiting times and high efficiency in service.

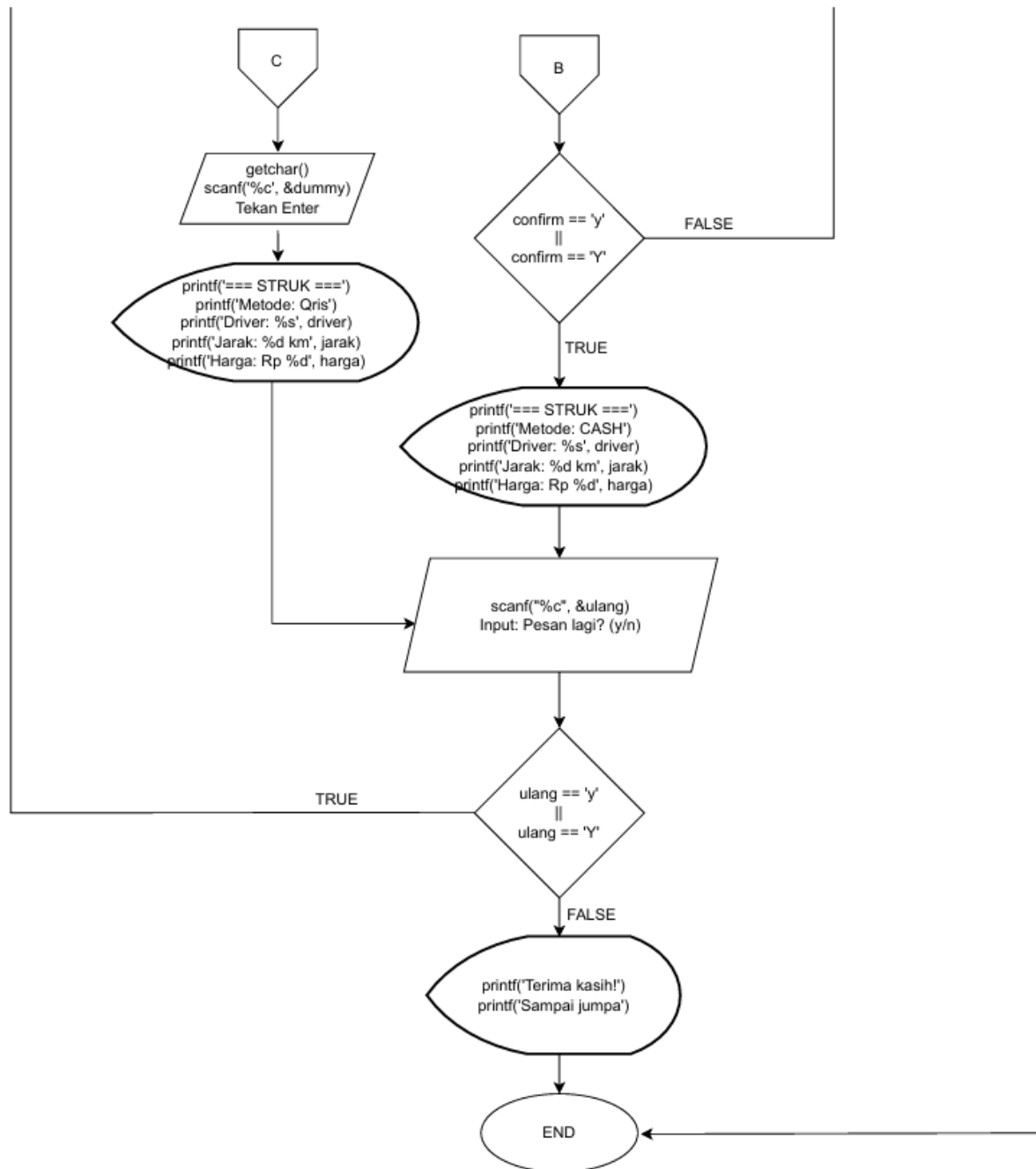
From the user's perspective, the GoRide system offers convenience through features such as fare estimates, driver tracking, and flexible payment methods, namely cash and QRIS. Meanwhile, for driver partners, the app provides the fastest route maps, daily income reports, and a customer rating system that helps maintain service quality.

Overall, the GoRide system relies on the integration of information technology, location data, and digital payment systems to create a fast, safe, and transparent transportation ecosystem. This analysis shows that GoRide's success lies in its ability to connect the mobility needs of urban communities with practical and easy-to-use technology.

FLOWCHART







SOURCE CODE

```
#include <stdio.h> // Library untuk input/output standar
#include <stdlib.h> // Library untuk fungsi umum seperti rand()
#include <string.h> // Library untuk manipulasi string
#include <time.h> // Library untuk waktu dan random seed

// Fungsi untuk mencetak header simulasi Go-Ride
void printHeader() {
    printf("=====\n");
    printf("          GO-RIDE (SIMULASI)          \n");
    printf("=====\n");
}

// Fungsi untuk menghitung harga otomatis berdasarkan jarak
// Tarif dasar Rp 5000, plus Rp 2000 per km
int hitungHarga(int jarakKm) {
    int tarifAwal = 5000; // tarif dasar
    int tarifPerKm = 2000; // tarif per km
    return tarifAwal + (tarifPerKm * jarakKm);
}

// Fungsi untuk mencetak struk pemesanan dengan detail lengkap
void printReceipt(char *from, char *to, char *driver, int jarak, int harga, char *metode) {
    time_t t = time(NULL); // Mendapatkan waktu saat ini
    struct tm tm = *localtime(&t); // Mengonversi ke format lokal
    printf("\n===== STRUK PEMESANAN GO-RIDE =====\n");
    printf("Waktu   : %04d-%02d-%02d %02d:%02d:%02d\n",
        tm.tm_year + 1900, tm.tm_mon + 1, tm.tm_mday,
        tm.tm_hour, tm.tm_min, tm.tm_sec);
```

```

printf("Rute   : %s -> %s\n", from, to);
printf("Driver : %s\n", driver);
printf("Jarak   : %d km\n", jarak);
printf("Harga   : Rp %d\n", harga);
printf("Bayar   : %s\n", metode);
printf("Status  : Berhasil\n");
printf("=====\\n\\n");
}

int main() {
    char from[50], to[50]; // Array untuk menyimpan lokasi penjemputan dan tujuan
    char driver[50];       // Array untuk menyimpan nama driver
    int jarak, harga, metode; // Variabel untuk jarak, harga, dan metode pembayaran
    int lagi = 1;          // Flag untuk loop ulang pemesanan

    srand(time(NULL)); // Inisialisasi random seed berdasarkan waktu saat ini, dilakukan sekali di awal

    while (lagi) { // Loop utama untuk simulasi pemesanan berulang
        printHeader(); // Mencetak header aplikasi

        printf("Masukkan lokasi penjemputan : ");
        scanf(" %[^\n]", from); // Input lokasi penjemputan, menggunakan %[^\n] untuk membaca spasi

        printf("Masukkan tujuan perjalanan  : ");
        scanf(" %[^\n]", to); // Input tujuan perjalanan

        // Simulasi jarak secara acak antara 3-30 km
        jarak = (rand() % 28) + 3;

        // Simulasi pemilihan driver secara acak dari daftar
        const char *drivers[] = {"Andi", "Budi", "Citra", "Dewi", "Eko", "Fajar", "Gilang"};
        strcpy(driver, drivers[rand() % 7]);
    }
}

```

```

// Hitung harga berdasarkan jarak menggunakan fungsi hitungHarga
harga = hitungHarga(jarak);

printf("\nDriver tersedia: %s\n", driver);
printf("Perkiraan jarak: %d km\n", jarak);
printf("Total harga   : Rp %d\n", harga);

printf("\nPilih metode pembayaran:\n");
printf("1) Cash\n");
printf("2) QRIS\n");
printf("Masukkan pilihan (1/2): ");
scanf("%d", &metode);
getchar(); // Membersihkan newline dari buffer input

if (metode == 1) { // Jika memilih Cash
    char confirm;
    printf("\nAnda memilih CASH.\n");
    printf("Konfirmasi bayar tunai kepada driver saat sampai? (y/n): ");
    confirm = getchar(); // Input konfirmasi
    getchar(); // Membersihkan buffer
    if (confirm == 'y' || confirm == 'Y') {
        printReceipt(from, to, driver, jarak, harga, "Cash"); // Cetak struk jika dikonfirmasi
    } else {
        printf("Pembayaran dibatalkan. Kembali ke menu.\n\n");
        continue; // Kembali ke awal loop
    }
}
} else if (metode == 2) { // Jika memilih QRIS
    printf("\nAnda memilih QRIS.\n");
    char qrisCode[64]; // Array untuk menyimpan kode QRIS simulasi
    time_t t = time(NULL); // Waktu saat ini untuk membuat kode unik
    snprintf(qrisCode, sizeof(qrisCode), "QRIS-%ld-%d", t % 100000, jarak); // Generate kode QRIS
    printf("Kode QRIS (silakan scan): %s\n", qrisCode);
}

```

```
printf("Tekan Enter setelah membayar via QRIS...");  
    getchar(); // Simulasi menunggu pembayaran  
    printReceipt(from, to, driver, jarak, harga, "QRIS"); // Cetak struk  
} else {  
    printf("Metode tidak valid. Kembali ke menu.\n\n");  
    continue; // Kembali ke awal loop jika input tidak valid  
}
```

```
printf("Pesan lagi? (y/n): ");  
char ulang = getchar(); // Input untuk ulang pemesanan  
getchar(); // Membersihkan buffer  
if (ulang == 'y' || ulang == 'Y') {  
    lagi = 1; // Lanjutkan loop  
    printf("\n");  
} else {  
    lagi = 0; // Akhiri loop  
    printf("\nTerima kasih telah menggunakan GoRide (simulasi)!\n");  
}  
}
```

```
return 0; // Mengakhiri program  
}
```

OUTPUT PROGRAM

1. Go-Ride program start screen

```
markdown
=====
GO-RIDE (SIMULASI)
=====
Masukkan lokasi penjemputan :
```

2. Enter the pickup location and destination in the Go-Ride program

```
yaml
Masukkan lokasi penjemputan : Kampus UI
Masukkan tujuan perjalanan  : Mall Depok
```

3. Driver information display and payment method selection on the Go-Ride app

```
yaml
Driver tersedia: Citra
Perkiraan jarak: 12 km
Total harga    : Rp 29000

Pilih metode pembayaran:
1) Cash
2) QRIS
Masukkan pilihan (1/2):
```

4. Payment display using the QRIS method on the Go-Ride program

```
java
Anda memilih QRIS.
Kode QRIS (silakan scan): QRIS-54321-12
Tekan Enter setelah membayar via QRIS...
```

5. Reorder confirmation display on the Go-Ride program

```
bash
```

```
Pesan lagi? (y/n):
```

6. Final display of the Go-Ride program

```
java
```

```
Terima kasih telah menggunakan GoRide (simulasi)!
```